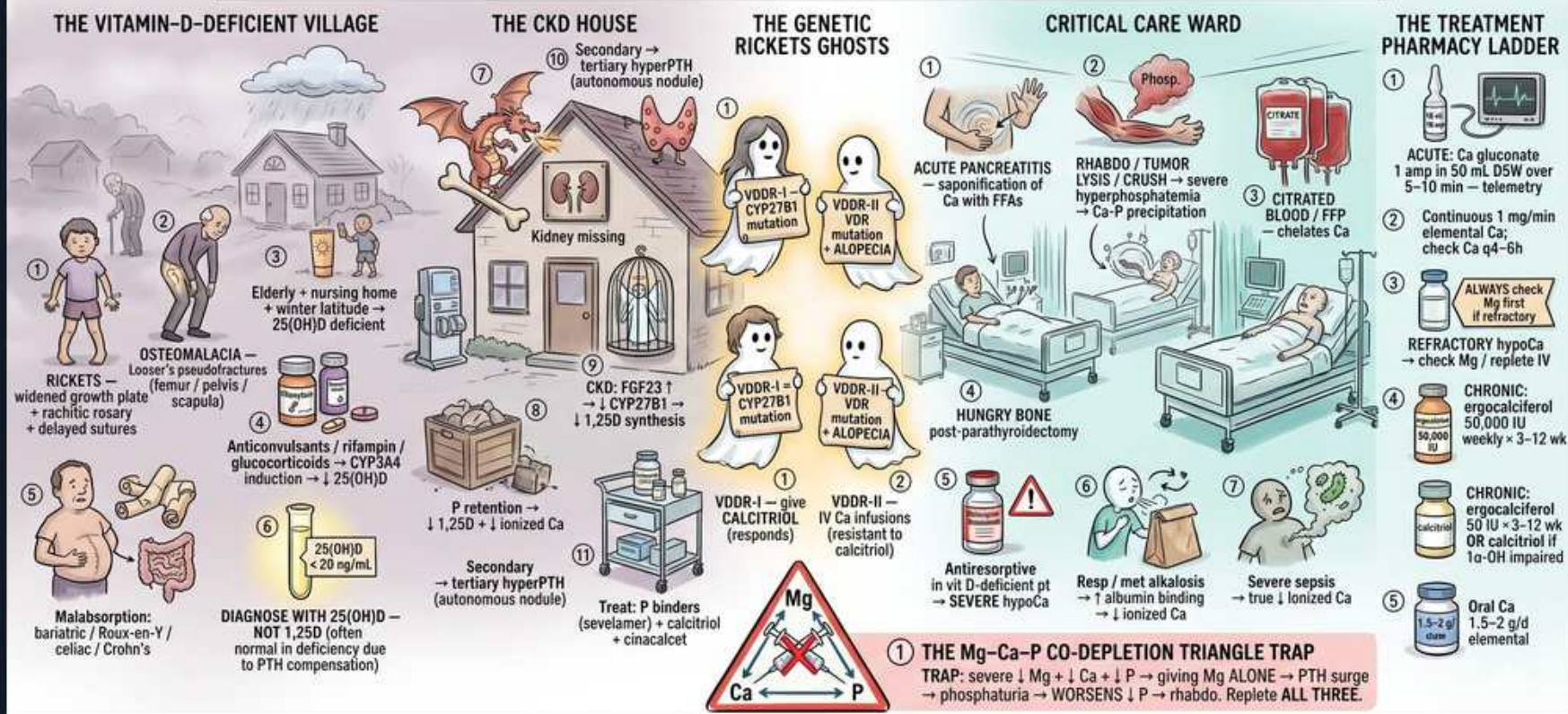


THE VIT-D-DEFICIENT VILLAGE & ICU — NON-PARATHYROID HYPOCALCEMIA & ACUTE CAUSES

Hypocalcemia from secondary hyperPTH (vit D / CKD) is usually Milder than from hypoPTH – parathyroids compensate at expense of bone



Diagnose vit D deficiency with 25(OH)D not 1,25D. CKD: 1FGF23 + retained P → ↓1,25D + secondary hyperPTH. Pancreatitis / rhabdo / TLS / citrate transfusion → acute hypoCa. Always check Mg in refractory hypocalcemia.

1. Secondary hyperPTH = milder

WHAT YOU SEE

Subtitle: hypocalcemia from secondary hyperPTH (vit D / CKD) is usually Milder than from hypoPTH — parathyroids compensate at the expense of bone

WHAT IT ENCODES

In NON-parathyroid hypocalcemia the glands are intact, so a falling Ca drives a compensatory rise in PTH (secondary hyperparathyroidism). This high PTH mobilizes Ca from bone, blunting the serum Ca drop — so hypocalcemia is usually milder/more chronic than in true hypoparathyroidism, but at the cost of skeletal demineralization (osteomalacia, renal osteodystrophy). The labs that separate the two camps: hypocalcemia with HIGH PTH = secondary (vit D deficiency, CKD); hypocalcemia with LOW/inappropriate PTH = hypoparathyroidism.

BOARD TRAP

WRONG: seeing low Ca + high PTH and diagnosing 'hyperparathyroidism.' That combination is APPROPRIATE secondary hyperPTH responding to hypocalcemia, not a primary gland problem — work up the cause of low Ca (check 25-OH-D, creatinine, magnesium, phosphate). Primary hyperPTH would show high Ca with high PTH, the opposite calcium.

2. Vit D deficiency -> osteomalacia

WHAT YOU SEE

Vitamin-D-Deficient Village: rickets / osteomalacia — Looser's zones, low 25(OH)D, secondary hyperPTH

WHAT IT ENCODES

Low 25(OH)D (sun deprivation, poor intake, malabsorption) -> reduced gut Ca AND phosphate absorption -> hypocalcemia -> secondary hyperPTH. The classic biochemical signature: LOW or low-normal Ca, LOW phosphate (PTH-driven phosphaturia), HIGH PTH, HIGH alkaline phosphatase, and LOW 25(OH)D (<20 ng/mL deficient). Chronically this causes osteomalacia in adults (Looser's zones/pseudofractures, bone pain, proximal myopathy) and rickets in children. Treat with cholecalciferol/ergocalciferol (e.g., 50,000 IU weekly x6-8 wk, then maintenance).

BOARD TRAP

WRONG: diagnosing vitamin D deficiency with a 1,25-OH-D (calcitriol) level. 1,25-D can be NORMAL or even HIGH in deficiency because secondary hyperPTH upregulates renal 1-alpha-hydroxylase. Always order 25(OH)D (the storage form) to diagnose deficiency. Also expect HIGH ALP and HIGH PTH — a normal PTH would be inappropriate.

3. Malabsorption causes

WHAT YOU SEE

Village: malabsorption — bariatric / Roux-en-Y / celiac / Crohn's

WHAT IT ENCODES

Fat malabsorption impairs absorption of fat-soluble vitamin D and binds luminal calcium to unabsorbed fatty acids. Causes: CELIAC disease, CROHN's/IBD, chronic pancreatitis, short-bowel, and BARIATRIC surgery (especially Roux-en-Y gastric bypass, which bypasses the duodenum/proximal jejunum where Ca and vitamin D are absorbed). Result: combined vitamin D + calcium deficiency, hypocalcemia, secondary hyperPTH, osteomalacia, and often low magnesium too.

BOARD TRAP

WRONG: treating a malabsorptive patient with standard ORAL vitamin D/calcium and expecting normalization. They malabsorb the replacement too — use higher doses, sometimes parenteral vitamin D, and co-replete magnesium. Also wrong to forget post-bariatric patients need lifelong Ca + vit D (and B12/iron) surveillance.

4. Drugs lowering vit D

WHAT YOU SEE

Village: anticonvulsants / rifampin / glucocorticoids -> induce CYP3A4 -> down 25(OH)D

WHAT IT ENCODES

Enzyme-inducing drugs accelerate vitamin D catabolism via hepatic CYP3A4/CYP24A1 induction, lowering 25(OH)D and active 1,25-D: classically PHENYTOIN, phenobarbital, carbamazepine (anticonvulsants), RIFAMPIN, and glucocorticoids (which also reduce gut Ca absorption and increase renal Ca loss). Over time this produces drug-induced osteomalacia/osteoporosis.

BOARD TRAP

WRONG: overlooking the medication list as the driver of low Ca/low vitamin D and bone loss. A patient on chronic phenytoin or long-term glucocorticoids needs proactive vitamin D + calcium repletion and bone-density monitoring; the fix is often dose adjustment plus supplementation rather than a hunt for occult malabsorption.

5. CKD: high FGF23, low 1,25-D

WHAT YOU SEE

CKD House: vit D / CKD; high FGF23 -> down CYP27B1 -> down 1,25-D synthesis; secondary -> tertiary hyperPTH

WHAT IT ENCODES

CKD-Mineral Bone Disorder: failing kidneys retain PHOSPHATE -> FGF23 rises (early marker) -> FGF23 suppresses renal 1-alpha-hydroxylase (CYP27B1) -> LOW 1,25-D -> reduced gut Ca absorption -> HYPOCALCEMIA. Hyperphosphatemia also directly binds Ca. The result is high phosphate, low/low-normal Ca, low 1,25-D, and progressively HIGH PTH (secondary hyperPTH). If parathyroids become autonomous over years -> tertiary hyperPTH (high PTH AND high Ca).

BOARD TRAP

WRONG: giving plain ergocalciferol/cholecalciferol alone for CKD hypocalcemia. The failing kidney cannot activate it (1-alpha-hydroxylase is suppressed) — you must give ACTIVE vitamin D (calcitriol or paricalcitol) PLUS phosphate binders to control phosphate first. Giving Ca/calcitriol without addressing high phosphate risks vascular/metastatic calcification.

6. Phosphate binders + calcitriol

WHAT YOU SEE

CKD House treatment: P binders (sevelamer) + calcitriol + cinacalcet

WHAT IT ENCODES

CKD-MBD management triad: (1) control phosphate with binders taken WITH meals — non-calcium binders (SEVELAMER, lanthanum) preferred, or calcium acetate/carbonate if Ca is low; (2) replace ACTIVE vitamin D (calcitriol/paricalcitol) to restore gut Ca absorption and suppress PTH; (3) CINACALCET (a calcimimetic that sensitizes the CaSR) to lower PTH without raising Ca/phosphate, useful in secondary hyperPTH on dialysis.

BOARD TRAP

WRONG: using cinacalcet in a hypocalcemic CKD patient without correcting Ca first. Cinacalcet LOWERS Ca (it activates CaSR -> suppresses PTH) and can precipitate symptomatic hypocalcemia — it is for high-PTH states with adequate/high Ca, not for fixing low Ca. Also avoid calcium-based binders if the patient is hypercalcemic or has vascular calcification.

7. Acute pancreatitis

WHAT YOU SEE

Critical Care Ward: ACUTE PANCREATITIS — saponification of fat, Ca-FFA precipitation

WHAT IT ENCODES

In acute pancreatitis, released pancreatic lipase liberates free fatty acids from peripancreatic and mesenteric fat; these FFAs chelate calcium (SAPONIFICATION) and deposit it as insoluble calcium soaps -> hypocalcemia. Glucagon-stimulated calcitonin and hypomagnesemia may also contribute. The degree of hypocalcemia tracks with severity.

BOARD TRAP

WRONG: ignoring hypocalcemia as incidental in pancreatitis. Hypocalcemia (Ca <8 mg/dL or ionized fall) is a marker of SEVERE disease and is a Ranson criterion (Ca <8 at 48 h) — it signals poor prognosis. Don't dismiss it; recognize it as a severity flag and check ionized Ca and magnesium.

8. Rhabdo / tumor lysis

WHAT YOU SEE

Critical Care: RHABDO / TUMOR / CRUSH — severe hyperphosphatemia -> Ca-PO₄ precipitation

WHAT IT ENCODES

Massive cell breakdown — RHABDOMYOLYSIS (crush/trauma) and TUMOR LYSIS SYNDROME (post-chemo of bulky/leukemia-lymphoma) — dumps intracellular phosphate into blood. The resulting hyperphosphatemia drives calcium-phosphate precipitation into tissues -> HYPOCALCEMIA (also hyperkalemia, hyperuricemia, high LDH, AKI). In rhabdo, Ca that precipitated in damaged muscle may be released LATER as a rebound hypercalcemia during recovery.

BOARD TRAP

WRONG: aggressively giving IV calcium to correct the lab hypocalcemia in tumor lysis/rhabdo. Unless the patient is SYMPTOMATIC (tetany, seizures, arrhythmia), withhold calcium — adding Ca to a high-phosphate milieu drives METASTATIC calcium-phosphate deposition (worsens kidney injury, soft-tissue calcification). Lower the phosphate first; treat the underlying process.

9. Citrate / blood products

WHAT YOU SEE

Critical Care: CITRATE -> stored blood / FFP -> chelates Ca

WHAT IT ENCODES

Stored blood products (PRBCs, FFP, platelets) and continuous renal replacement therapy use CITRATE as anticoagulant. Citrate chelates ionized calcium; with massive transfusion or impaired hepatic citrate metabolism (liver disease, hypothermia, shock), ionized Ca falls -> hypocalcemia (and can cause QT prolongation, hypotension). Total Ca may look normal because the chelated Ca is still 'counted.'

BOARD TRAP

WRONG: trusting TOTAL calcium during massive transfusion or citrate CRRT. Citrate binds the IONIZED (physiologically active) fraction, so total Ca can be misleadingly normal while ionized Ca is dangerously low — always monitor IONIZED Ca and give calcium gluconate/chloride to keep it in range.

10. Hungry bone (post-PTHx)

WHAT YOU SEE

Critical Care: HUNGRY BONE — post-parathyroidectomy

WHAT IT ENCODES

Hungry bone syndrome occurs after PARATHYROIDECTOMY (or thyroidectomy, or correction of severe hyperthyroidism) when chronically high PTH is abruptly removed: the previously resorbing skeleton avidly re-mineralizes, rapidly pulling Ca, PHOSPHATE, and MAGNESIUM out of the blood -> profound, prolonged hypocalcemia (can last days to weeks). Risk highest with high preop PTH/ALP and large bone disease.

BOARD TRAP

WRONG: expecting transient, easily-corrected hypocalcemia after parathyroid surgery. Hungry bone causes SEVERE, sustained hypocalcemia (low Ca, low PO₄, low Mg) needing aggressive IV/oral calcium PLUS calcitriol PLUS magnesium repletion over days. Distinguish it from surgical HYPOPARATHYROIDISM (low PTH) — hungry bone has low PO₄ whereas hypoPTH has HIGH phosphate.

11. Hypomagnesemia triangle

WHAT YOU SEE

Mg-Ca-PO₄ Co-Depletion Triangle Trap: low Mg -> low PTH + PTH resistance; refractory hypocalcemia

WHAT IT ENCODES

Severe hypomagnesemia (Mg <1 mg/dL) causes hypocalcemia by a DOUBLE hit: it impairs PTH SECRETION and also induces end-organ PTH RESISTANCE -> functional hypoparathyroidism with an inappropriately low/normal PTH despite low Ca. Causes of low Mg: alcoholism, diarrhea/malabsorption, loop diuretics, PPIs, aminoglycosides, cisplatin, Gitelman. The three often fall together (Mg, Ca, PO₄), so check all three.

BOARD TRAP

WRONG: pouring in more calcium when hypocalcemia 'won't correct' despite repletion. If Ca is refractory, CHECK AND REPLETE MAGNESIUM FIRST — until Mg is restored, PTH cannot be secreted or act, so Ca will not rise no matter how much you give. Refractory hypocalcemia = think magnesium.

12. IV calcium gluconate (acute)

WHAT YOU SEE

Treatment Pharmacy Ladder: ACUTE — Ca gluconate 1-2 g IV in D5W over 5-10 min -> telemetry

WHAT IT ENCODES

Acute SYMPTOMATIC hypocalcemia (tetany, Chvostek/Trousseau signs, laryngospasm, seizures, QT prolongation/arrhythmia) is treated with IV CALCIUM GLUCONATE 1-2 g in D5W over 5-10 min, then a continuous infusion, with cardiac/telemetry monitoring. Calcium gluconate is preferred peripherally (calcium chloride is more concentrated/irritating, reserved for central lines/codes). Transition to oral calcium + calcitriol once stable.

BOARD TRAP

WRONG: giving IV calcium without simultaneously checking/correcting MAGNESIUM and being cautious of the patient's DIGOXIN status — rapid IV calcium can precipitate fatal digoxin toxicity ('stone heart'). And in hyperphosphatemic states (tumor lysis/rhabdo) reserve IV Ca for symptomatic patients only, to avoid metastatic calcification.